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Grade	**78.83** out of 100.00

Question 1

Correct

Mark 1.00 out of 1.00

What is 1 - 0.1 - 0.1 - 0.1 - 0.1 - 0.1 == 0.5?

Select one:

- A. True
- B. False
- C. There is no guarantee that 1 - 0.1 - 0.1 - 0.1 - 0.1 - 0.1 == 0.5 is true. ✔ See Common Error 3 in Section 2.19. This epression involves floating-point number. Floating-point numbers are approximated. The correct answer is C.

See Common Error 3 in Section 2.19. This epression involves floating-point number. Floating-point numbers are approximated. The correct answer is C.

See Common Error 3 in Section 2.19. This epression involves floating-point number. Floating-point numbers are approximated. The correct answer is C.

The correct answer is: There is no guarantee that 1 - 0.1 - 0.1 - 0.1 - 0.1 - 0.1 == 0.5 is true.

Question 2

Correct

Mark 1.00 out of 1.00

Which of the following statements is false?

Select one:

- A. (x > 0 && x < 10) is same as ((x > 0) && (x < 10))
- B. (x > 0 || x < 10) is same as ((x > 0) || (x < 10))
- C. (x > 0 || x < 10 && y < 0) is same as (x > 0 || (x < 10 && y < 0))
- D. (x > 0 || x < 10 && y < 0) is same as ((x > 0 || x < 10) && y < 0) ✔ In D, && is evaluated before the || operator. So (x > 0 || x < 10 && y < 0) is not same as ((x > 0 || x < 10) && y < 0).

In D, && is evaluated before the || operator. So (x > 0 || x < 10 && y < 0) is not same as ((x > 0 || x < 10) && y < 0).

In D, && is evaluated before the || operator. So (x > 0 || x < 10 && y < 0) is not same as ((x > 0 || x < 10) && y < 0).

The correct answer is: (x > 0 || x < 10 && y < 0) is same as ((x > 0 || x < 10) && y < 0)

Question 3

Correct

Mark 2.00 out of 2.00

Analyze the following code:

```
if (x < 100) && (x > 10)
System.out.println("x is between 10 and 100");
```

Select one:

- A. The statement has compile errors because (x<100) & (x > 10) must be enclosed inside parentheses. ✔ The condition for an if statement must be enclosed in the parentheses. The correct answer is A.
- B. The statement has compile errors because (x<100) & (x > 10) must be enclosed inside parentheses and the println(...) statement must be put inside a block.
- C. The statement compiles fine.
- D. The statement compiles fine, but has a runtime error.

The condition for an if statement must be enclosed in the parentheses. The correct answer is A.

The condition for an if statement must be enclosed in the parentheses. The correct answer is A.

The correct answer is: The statement has compile errors because (x<100) & (x > 10) must be enclosed inside parentheses.

Question 4

Correct

Mark 1.00 out of 1.00

The _____ method returns a raised to the power of b.

Select one:

- A. Math.power(a, b)
- B. Math.pow(b, a)
- C. Math.exponent(a, b)
- D. Math.pow(a, b) ✔ See line 1 in Section 2.9.3.

See line 1 in Section 2.9.3.

See line 1 in Section 2.9.3.

The correct answer is: Math.pow(a, b)

Question 5

Correct

Mark 1.00 out of 1.00

What is the output from System.out.println((int)Math.random() * 4)?

Select one:

- A. 0 ✔ Casting is performed before the * operator in (int)Math.random() * 4. So, it returns 0.
- B. 1
- C. 2
- D. 3
- E. 4

Casting is performed before the * operator in (int)Math.random() * 4. So, it returns 0.

Casting is performed before the * operator in (int)Math.random() * 4. So, it returns 0.

The correct answer is: 0

Question 6

Correct

Mark 1.00 out of 1.00

If you enter 1 2 3, when you run this program, what will be the output?

```
import java.util.Scanner;

public class Test1 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter three numbers: ");
    double number1 = input.nextDouble();
    double number2 = input.nextDouble();
    double number3 = input.nextDouble();

    // Compute average
    double average = (number1 + number2 + number3) / 3;

    // Display result
    System.out.println(average);
  }
}
```

Select one:

- A. 1.0
- B. 2.0 ✔ (1.0 + 2.0 + 3.0) / 3 is 2.0
- C. 3.0
- D. 4.0

(1.0 + 2.0 + 3.0) / 3 is 2.0

(1.0 + 2.0 + 3.0) / 3 is 2.0

The correct answer is: 2.0

Question 7

Correct

Mark 2.00 out of 2.00

What is y after the following switch statement is executed?

```
int x = 3; int y = 4;
switch (x + 3) {
  case 6 -> y = 0;
  case 7 -> y = 1;
  default -> y += 1;
}
```

Select one:

- A. 1
- B. 2
- C. 3
- D. 4
- E. 0 ✔ Since x is 3, x + 3 is 6. So, case 6 is executed. y is 0. Since the -> operator is used here. The switch statement is now finished. After the switch statement, y is 0.

Since x is 3, x + 3 is 6. So, case 6 is executed. y is 0. Since the -> operator is used here. The switch statement is now finished. After the switch statement, y is 0.

Since x is 3, x + 3 is 6. So, case 6 is executed. y is 0. Since the -> operator is used here. The switch statement is now finished. After the switch statement, y is 0.

The correct answer is: 0

Question 8

Correct

Mark 1.00 out of 1.00

Which of the following are correct names for variables according to Java naming conventions? Please select all that apply.

Select one or more:

- A. FindArea
- B. RADIUS
- C. findArea ✔ A single-word variable is in lowercase. In a multiple-word variable, the words are concatenated with the first word in lowercase and the first letter of each subsequent word in uppercase.
- D. Radius
- E. radius ✔ A single-word variable is in lowercase. In a multiple-word variable, the words are concatenated with the first word in lowercase and the first letter of each subsequent word in uppercase.

A single-word variable is in lowercase. In a multiple-word variable, the words are concatenated with the first word in lowercase and the first letter of each subsequent word in uppercase.

A single-word variable is in lowercase. In a multiple-word variable, the words are concatenated with the first word in lowercase and the first letter of each subsequent word in uppercase.

The correct answers are: radius, findArea

Question 9

Correct

Mark 1.00 out of 1.00

The following code fragment reads in two numbers. What is the incorrect way to enter these two numbers?

```
Scanner input = new Scanner(System.in);
int i = input.nextInt();
double d = input.nextDouble();
```

Select one:

- A. Enter an integer, an Enter key, a double value, and then the Enter key.
- B. Enter an integer, a space, a double value, and then the Enter key.
- C. Enter an integer, two spaces, a double value, and then the Enter key.
- D. Enter a numeric value with a decimal point, a space, an integer, and then the Enter key. ✔ See Listing 2.3.

See Listing 2.3.

See Listing 2.3.

The correct answer is: Enter a numeric value with a decimal point, a space, an integer, and then the Enter key.

Question 10

Correct

Mark 1.00 out of 1.00

What is y displayed?

```
public class Test {
  public static void main(String[] args) {
    int x = 1;
    int y = x + x++;
    System.out.println("y is " + y);
  }
}
```

Select one:

- A. y is 1.
- B. y is 2. ✔ When evaluating x + x++, x is evaluated first, which is 1. X++ returns 1 since it is post-increment and 2. Therefore y is 1 + 1.
- C. y is 3.
- D. y is 4.

When evaluating x + x++, x is evaluated first, which is 1. X++ returns 1 since it is post-increment and 2. Therefore y is 1 + 1.

When evaluating x + x++, x is evaluated first, which is 1. X++ returns 1 since it is post-increment and 2. Therefore y is 1 + 1.

The correct answer is: y is 2.

Question 11

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
boolean even = false;
System.out.println(even ? "true" : "false");
```

Select one:

- A. true
- B. false ✔ Since even is false, the conditional expression yields false. The correct answer is B.
- C. nothing
- D. true false

Since even is false, the conditional expression yields false. The correct answer is B.

Since even is false, the conditional expression yields false. The correct answer is B.

The correct answer is: false

Question 12

Correct

Mark 1.00 out of 1.00

Which of the following statements are the same?

(A) x -= x + 4
(B) x = x + 4 - x
(C) x = x - (x + 4)

Select one:

- A. (A) and (B) are the same.
- B. (A) and (C) are the same. ✔ See Table 2.4.
- C. (B) and (C) are the same.
- D. (A), (B), and (C) are the same.

See Table 2.4.

See Table 2.4.

The correct answer is: (A) and (C) are the same.

Question 13

Incorrect

Mark 0.00 out of 2.00

Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    int n = 10000 * 10000 * 10000;
    System.out.println("n is " + n);
  }
}
```

Select one:

- A. The result of 10000 * 10000 * 10000 is too large to be stored in an int variable n. This causes an overflow and the program is aborted. ✖
- B. The program displays n is 1000000000000.
- C. The result of 10000 * 10000 * 10000 is too large to be stored in an int variable n. This causes an underflow and the program is aborted.
- D. The result of 10000 * 10000 * 10000 is too large to be stored in an int variable n. This causes an overflow and the program continues to execute because Java does not report errors on overflow.
- E. The result of 10000 * 10000 * 10000 is too large to be stored in an int variable n. This causes an underflow and the program continues to execute because Java does not report errors on underflow.

See Common Error 2: Integer Overflow.

The correct answer is: The result of 10000 * 10000 * 10000 is too large to be stored in an int variable n. This causes an overflow and the program continues to execute because Java does not report errors on overflow.

Question 14

Correct

Mark 2.00 out of 2.00

Assume x = 4 and y = 5, which of the following is true?

Select one:

- A. !(x == 4) ^ y != 5
- B. x != 4 ^ y == 5 ✔ x != 4 is false and y == 5 is true. So B is correct.
- C. x == 5 ^ y == 4
- D. x != 5 ^ y != 4

x != 4 is false and y == 5 is true. So B is correct.

x != 4 is false and y == 5 is true. So B is correct.

The correct answer is: x != 4 ^ y == 5

Question 15

Correct

Mark 2.00 out of 2.00

To obtain the current second, use _____.

Select one:

- A. System.currentTimeMillis() % 60
- B. System.currentTimeMillis() / 1000 / 60 % 60
- C. System.currentTimeMillis() % 3600
- D. System.currentTimeMillis() / 1000 % 60 ✔ See Listing 2.7.
- E. System.currentTimeMillis() / 1000 / 60 / 60 % 24

See Listing 2.7.

See Listing 2.7.

The correct answer is: System.currentTimeMillis() / 1000 % 60

Question 16

Correct

Mark 2.00 out of 2.00

The expression (int)(76.0252175 * 100) / 100 evaluates to _____.

Select one:

- A. 76 ✔ In order to obtain 76.02, you have to divide 100.0.
- B. 76.0252175
- C. 76.02
- D. 76.03

In order to obtain 76.02, you have to divide 100.0.

In order to obtain 76.02, you have to divide 100.0.

The correct answer is: 76

Question 17

Incorrect

Mark 0.00 out of 2.00

What is i printed?

```
public class Test {
  public static void main(String[] args) {
    int j = 0;
    int i = ++j + j * 5;

    System.out.println("What is i? " + i);
  }
}
```

Select one:

- A. 0
- B. 1 ✖
- C. 5
- D. 6

Operands are evaluated from left to right in Java. The left-hand operand of a binary operator is evaluated before any part of the right-hand operand is evaluated. This rule takes precedence over any other rules that govern expressions. Therefore, ++j is evaluated first, and j is now 1. Then j * 5 is evaluated, returns 5. So, i is 6.

The correct answer is: 6

Question 18

Correct

Mark 2.00 out of 2.00

Which of these data types requires the most amount of memory?

Select one:

- A. long ✔ long takes 8 bytes. int 4 bytes. short 2 bytes. byte 1 byte.
- B. byte
- C. short
- D. int

long takes 8 bytes. int 4 bytes. short 2 bytes. byte 1 byte.

long takes 8 bytes. int 4 bytes. short 2 bytes. byte 1 byte.

The correct answer is: long

Question 19

Incorrect

Mark 0.00 out of 1.00

Math.pow(4, 1 / 2) returns _____.

Select one:

- A. 1.0
- B. 0
- C. 1
- D. 2.0
- E. 2 ✖

Note that 1 / 2 is 0.

The correct answer is: 1.0

Question 20

Correct

Mark 1.00 out of 1.00

The expression 4 + 20 / (3 - 1) * 2 is evaluated to _____.

Select one:

- A. 24 ✔ See the second paragraph in this section.
- B. 25
- C. 20
- D. 9
- E. 4

See the second paragraph in this section.

See the second paragraph in this section.

The correct answer is: 24

Question 21

Incorrect

Mark 0.00 out of 1.00

What is y displayed in the following code?

```
public class Test {
  public static void main(String[] args) {
    int x = 1;
    int y = x++ + x;
    System.out.println("y is " + y);
  }
}
```

Select one:

- A. y is 1.
- B. y is 2. ✖
- C. y is 3.
- D. y is 4.

When evaluating x++ + x, x++ is evaluated first, which does two things: 1. returns 1 since it is post-increment. x becomes 2. Therefore y is 1 + 2.

The correct answer is: y is 3.

Question 22

Correct

Mark 1.00 out of 1.00

Analyze the following code:

Code 1:

```
int number = 45;
boolean even;

if (number % 2 == 0)
  even = true;
else
  even = false;
```

Code 2:

```
int number = 45;
boolean even = (number % 2 == 0);
```

Select one:

- A. Code 1 has compile errors.
- B. Code 2 has compile errors.
- C. Both Code 1 and Code 2 have compile errors.
- D. Both Code 1 and Code 2 are correct, but Code 2 is better. ✔ Both Code 1 and Code 2 are correct. Clearly Code 2 is shorter and better.

Both Code 1 and Code 2 are correct. Clearly Code 2 is shorter and better.

Both Code 1 and Code 2 are correct. Clearly Code 2 is shorter and better.

The correct answer is: Both Code 1 and Code 2 are correct, but Code 2 is better.

Question 23

Correct

Mark 1.00 out of 1.00

What is the value of (double)5/2?

Select one:

- A. 2
- B. 3.0
- C. 3
- D. 2.5 ✔ See the second code box in this section.
- E. 2.0

See the second code box in this section.

See the second code box in this section.

The correct answer is: 2.5

Question 24

Correct

Mark 1.00 out of 1.00

Suppose isPrime is a boolean variable, which of the following is the correct and best statement for testing if isPrime is true?

Select one:

- A. if (isPrime = true)
- B. if (isPrime == true)
- C. if (isPrime) ✔ A is wrong. It assigns true to isPrime. B is correct, but C is better. D is wrong. It assigns false to isPrime. E is correct, but C is clearly better.
- D. if (!isPrime = false)
- E. if (!isPrime == false)

A is wrong. It assigns true to isPrime. B is correct, but C is better. D is wrong. It assigns false to isPrime. E is correct, but C is clearly better.

A is wrong. It assigns true to isPrime. B is correct, but C is better. D is wrong. It assigns false to isPrime. E is correct, but C is clearly better.

The correct answer is: if (isPrime)

Question 25

Correct

Mark 2.00 out of 2.00

The System.currentTimeMillis() returns _____.

Select one:

- A. the current time in milliseconds since midnight, January 1, 1970
- B. the current time
- C. the current time in milliseconds since midnight, January 1, 1970 GMT (the Unix time) ✔ See Listing 2.7.
- D. the current time in milliseconds
- E. the current time in milliseconds since midnight

See Listing 2.7.

See Listing 2.7.

The correct answer is: the current time in milliseconds since midnight, January 1, 1970 GMT (the Unix time)

Question 26

Correct

Mark 1.00 out of 1.00

What is x after the following statements?

```
int x = 1;
x *= x + 1;
```

Select one:

- A. x is 1.
- B. x is 2. ✔ See Table 2.4.
- C. x is 3.
- D. x is 4.

See Table 2.4.

See Table 2.4.

The correct answer is:
x is 2.

Question 27

Correct

Mark 2.00 out of 2.00

What is the exact output of the following code?

```
double area = 3.5;
System.out.print("area");
System.out.print(area);
```

Select one:

- A. 3.5 3.5
- B. area 3.5
- C. 3.53.5
- D. area3.5 ✔ The first print statement prints a string followed by the second print statement that prints a number.

The first print statement prints a string followed by the second print statement that prints a number.

The first print statement prints a string followed by the second print statement that prints a number.

The correct answer is: area3.5

Question 28

Correct

Mark 2.00 out of 2.00

What is the result of 45 / 4?

Select one:

- A. 11.25
- B. 10
- C. 12
- D. 11 ✔ The result of the division is the quotient and the fractional part is truncated. So, 45 / 4 is 11. 3 / 2 is 1, and so on.

The result of the division is the quotient and the fractional part is truncated. So, 45 / 4 is 11. 3 / 2 is 1, and so on.

The result of the division is the quotient and the fractional part is truncated. So, 45 / 4 is 11. 3 / 2 is 1, and so on.

The correct answer is: 11

Question 29

Correct

Mark 1.00 out of 1.00

In Java, the word true is _____.

Select one:

- A. a Java keyword
- B. same as value 1
- C. a Boolean literal ✔ true is a Boolean literal just like integer literal 10.
- D. same as value 0

true is a Boolean literal just like integer literal 10.

true is a Boolean literal just like integer literal 10.

The correct answer is: a Boolean literal

Question 30

Correct

Mark 2.00 out of 2.00

Suppose int x = 3264, what is the output of the following code?

```
int y = x % 10;
x = x / 10;
System.out.println("x is " + x + " and y is " + y);
```

Select one:

- A. x is 326 and y is 4 ✔ 3264 / 10 is 326 and 3264 % 10 is 4.
- B. x is 3264 and y is 326.4
- C. x is 326 and y is 326
- D. x is 3264 and y is 4
- E. x is 4 and y is 326

3264 / 10 is 326 and 3264 % 10 is 4.

3264 / 10 is 326 and 3264 % 10 is 4.

The correct answer is: x is 326 and y is 4

Question 31

Correct

Mark 2.00 out of 2.00

Which of the following are the same as 1545.534? Please select all that apply.

Select one or more:

- A. 1.545534e+3 ✔ See Section 2.10.3.
- B. 1545534.0e-3 ✔ See Section 2.10.3.
- C. 0.1545534e+4 ✔ See Section 2.10.3.
- D. 154553.4e-2 ✔ See Section 2.10.3.

See Section 2.10.3.

See Section 2.10.3.

The correct answers are: 1.545534e+3, 0.1545534e+4, 1545534.0e-3, 154553.4e-2

Question 32

Correct

Mark 1.00 out of 1.00

Which of the following are correct ways to declare variables? Please select all that apply.

Select one or more:

- A. int length; int width; ✔ Note that a semicolon ends a statement. In B, length and width are both declared as int.
- B. int length, width; ✔ Note that a semicolon ends a statement. In B, length and width are both declared as int.
- C. int length; width;
- D. int length, int width;

Note that a semicolon ends a statement. In B, length and width are both declared as int.

Note that a semicolon ends a statement. In B, length and width are both declared as int.

The correct answers are: int length; int width;, int length, width;

Question 33

Incorrect

Mark 0.00 out of 2.00

If you attempt to add an int, a byte, a long, and a double, the result will be a(n) _____ value.

Select one:

- A. double
- B. long ✖
- C. byte
- D. int

See the second paragraph in this section.

The correct answer is: double

Question 34

Correct

Mark 2.00 out of 2.00

Suppose a Scanner object is created as follows, what method do you use to read a real number?

Scanner input = new Scanner(System.in);

Select one:

- A. input.double();
- B. input.nextdouble();
- C. input.nextDouble(); ✔ The correct method to read a real number is nextDouble().
- D. input.Double();

The correct method to read a real number is nextDouble().

The correct method to read a real number is nextDouble().

The correct answer is: input.nextDouble();

Question 35

Correct

Mark 1.00 out of 1.00

Which of the following is equivalent to x != y? Please select all that apply.

Select one or more:

- A. ! (x == y) ✔ x != y means !(x == y) and x > y || x < y.
- B. x > y || x < y ✔ x != y means !(x == y) and x > y || x < y.
- C. x >= y || x <= y
- D. x > y && x < y

x != y means !(x == y) and x > y || x < y.

x != y means !(x == y) and x > y || x < y.

The correct answers are: ! (x == y), x > y || x < y

Question 36

Incorrect

Mark 0.00 out of 2.00

What is y after the following switch statement is executed?

```
int x = 3; int y = 4;
switch (x + 3) {
  case 6: y = 0;
  case 7: y = 1;
  default: y += 1;
}
```

Select one:

- A. 1
- B. 2
- C. 3
- D. 4
- E. 0 ✖

Since x is 3, x + 3 is 6. So, case 6 is executed. Since there is no break statement, the statement in the next case is executed. y is now 1. Finally y += 1 adds 1 to y. So y is 2. The correct answer is B.

The correct answer is: 2

Question 37

Correct

Mark 1.00 out of 1.00

To assign a value 1 to variable x, you write _____.

Select one:

- A. 1 = x;
- B. 1 := x;
- C. x == 1;
- D. x := 1;
- E. x = 1; ✔ See the first paragraph.

See the first paragraph.

See the first paragraph.

The correct answer is: x = 1;

Question 38

Correct

Mark 2.00 out of 2.00

Analyze the following program fragment:

```
int x;
double d = 1.5;

switch (d) {
  case 1.0: x = 1;
  case 1.5: x = 2;
  case 2.0: x = 3;
}
```

Select one:

- A. The program has a compile error because the required break statement is missing in the switch statement.
- B. The program has a compile error because the required default case is missing in the switch statement.
- C. The switch control variable cannot be double. ✔ The switch value cannot be a floating-point number. So the correct answer is C.
- D. No errors.

The switch value cannot be a floating-point number. So the correct answer is C.

The switch value cannot be a floating-point number. So the correct answer is C.

The correct answer is: The switch control variable cannot be double.

Question 39

Incorrect

Mark 0.00 out of 1.00

The _____ method immediately terminates the program.

Select one:

- A. System.exit(0);
- B. System.terminate(0); ✖
- C. System.halt(0);
- D. System.stop(0);
- E. System.quit(0);

System.exit(0) method can be used to terminate a program.

The correct answer is: System.exit(0);

Question 40

Incorrect

Mark 0.00 out of 1.00

To add number to sum, you write _____. (Note: Java is case-sensitive) Please select all that apply.

Select one or more:

- A. number = sum + number; ✖
- B. sum = sum + number;
- C. sum += number;
- D. sum = Number + sum;
- E. number += sum; ✖

See Table 2.4.

The correct answers are: sum += number;, sum = sum + number;

Question 41

Correct

Mark 1.00 out of 1.00

What is the possible output from System.out.println((int)(Math.random() * 4))? Please select all that apply.

Select one or more:

- A. 0 ✔ Math.random() returns a real value between 0.0 and 1.0, excluding 1.0. Math.random() * 4 yields a real value between 0.0 and 4.0, excluding 4.0. After casting, the resulting integer may be 0, 1, 2, or 3.
- B. 1 ✔ Math.random() returns a real value between 0.0 and 1.0, excluding 1.0. Math.random() * 4 yields a real value between 0.0 and 4.0, excluding 4.0. After casting, the resulting integer may be 0, 1, 2, or 3.
- C. 2 ✔ Math.random() returns a real value between 0.0 and 1.0, excluding 1.0. Math.random() * 4 yields a real value between 0.0 and 4.0, excluding 4.0. After casting, the resulting integer may be 0, 1, 2, or 3.
- D. 3 ✔ Math.random() returns a real value between 0.0 and 1.0, excluding 1.0. Math.random() * 4 yields a real value between 0.0 and 4.0, excluding 4.0. After casting, the resulting integer may be 0, 1, 2, or 3.
- E. 4

Math.random() returns a real value between 0.0 and 1.0, excluding 1.0. Math.random() * 4 yields a real value between 0.0 and 4.0, excluding 4.0. After casting, the resulting integer may be 0, 1, 2, or 3.

Math.random() returns a real value between 0.0 and 1.0, excluding 1.0. Math.random() * 4 yields a real value between 0.0 and 4.0, excluding 4.0. After casting, the resulting integer may be 0, 1, 2, or 3.

The correct answers are: 0, 1, 2, 3

Question 42

Partially correct

Mark 0.50 out of 1.00

Which of the following is a valid identifier? Please select all that apply.

Select one or more:

- A. $343
- B. 8+9
- C. class ✖
- D. radius ✔ class is a keyword, which cannot be used as an identifier. Identifiers cannot start with a number.
- E. 9X

class is a keyword, which cannot be used as an identifier. Identifiers cannot start with a number.

The correct answers are: $343, radius

Question 43

Correct

Mark 1.00 out of 1.00

To declare a constant MAX_LENGTH inside a method with value 99.98, you write _____.

Select one:

- A. double MAX_LENGTH = 99.98;
- B. final int MAX_LENGTH = 99.98;
- C. final double MAX_LENGTH = 99.98; ✔ See the first paragraph.
- D. final MAX_LENGTH = 99.98;

See the first paragraph.

See the first paragraph.

The correct answer is: final double MAX_LENGTH = 99.98;

Question 44

Correct

Mark 1.00 out of 1.00

Which of the following is the correct expression that evaluates to true if the number x is between 1 and 100 or the number is negative?

Select one:

- A. 1 < x < 100 && x < 0
- B. ((x < 100) && (x > 1)) || (x < 0) ✔ A and D have syntax errors. B uses || for the OR operator. The correct answer is B.
- C. ((x < 100) && (x > 1)) && (x < 0)
- D. (1 > x > 100) || (x < 0)

A and D have syntax errors. B uses || for the OR operator. The correct answer is B.

A and D have syntax errors. B uses || for the OR operator. The correct answer is B.

The correct answer is: ((x < 100) && (x > 1)) || (x < 0)

Question 45

Correct

Mark 1.00 out of 1.00

To add a value 1 to variable x, you write _____. Please select all that apply.

Select one or more:

- A. x += 1; ✔ See Table 2.4.
- B. x = x + 1; ✔ See Table 2.4.
- C. x = 1 + x; ✔ See Table 2.4.
- D. x := 1;
- E. 1 + x = x;

See Table 2.4.

See Table 2.4.

The correct answers are: x += 1;, x = x + 1;, x = 1 + x;

Question 46

Correct

Mark 2.00 out of 2.00

Suppose x=10 and y=10. What is x after evaluating the expression (y >= 10) || (x++ > 10)?

Select one:

- A. 9
- B. 10 ✔ For the || operator, the right operand is not evaluated, if the left operand is evaluated as true.
- C. 11

For the || operator, the right operand is not evaluated, if the left operand is evaluated as true.

For the || operator, the right operand is not evaluated, if the left operand is evaluated as true.

The correct answer is: 10

Question 47

Correct

Mark 1.00 out of 1.00

Analyze the following code.

```
boolean even = false;
if (even) {
  System.out.println("It is even!");
}
```

Select one:

- A. The code displays It is even!
- B. The code displays nothing. ✔ Since even is false, the if statement body is not executed. So, the correct answer is B.
- C. The code is wrong. You should replace if (even) with if (even == true).
- D. The code is wrong. You should replace if (even) with if (even = true).

Since even is false, the if statement body is not executed. So, the correct answer is B.

Since even is false, the if statement body is not executed. So, the correct answer is B.

The correct answer is: The code displays nothing.

Question 48

Correct

Mark 1.00 out of 1.00

What is 1 + 1 + 1 + 1 + 1 == 5?

Select one:

- A. True ✔ These are all integers. Integer arithmetic is accurate. So 1 + 1 + 1 + 1 + 1 is exactly 5.
- B. False
- C. There is no guarantee that 1 + 1 + 1 + 1 + 1 == 5 is true.

These are all integers. Integer arithmetic is accurate. So 1 + 1 + 1 + 1 + 1 is exactly 5.

These are all integers. Integer arithmetic is accurate. So 1 + 1 + 1 + 1 + 1 is exactly 5.

The correct answer is: True

Question 49

Correct

Mark 1.00 out of 1.00

Suppose x=10 and y=10. What is x after evaluating the expression (y > 10) && (x-- > 10)?

Select one:

- A. 9
- B. 10 ✔ For the && operator, the right operand is not evaluated, if the left operand is evaluated as false.
- C. 11

For the && operator, the right operand is not evaluated, if the left operand is evaluated as false.

For the && operator, the right operand is not evaluated, if the left operand is evaluated as false.

The correct answer is: 10

Question 50

Correct

Mark 2.00 out of 2.00

Analyze the following code.

```
public class Test {
  public static void main(String[] args) {
    int month = 09;
    System.out.println("month is " + month);
  }
}
```

Select one:

- A. The program has a syntax error, because 09 is an incorrect literal value. ✔ Any numeric literal with the prefix 0 is an octal value. But 9 is not an octal digit. An octal digit is 0, 1, 2, 3, 4, 5, 6, or 7.
- B. The program displays month is 9.
- C. The program displays month is 9.0.
- D. The program displays month is 09.

Any numeric literal with the prefix 0 is an octal value. But 9 is not an octal digit. An octal digit is 0, 1, 2, 3, 4, 5, 6, or 7.

Any numeric literal with the prefix 0 is an octal value. But 9 is not an octal digit. An octal digit is 0, 1, 2, 3, 4, 5, 6, or 7.

The correct answer is: The program has a syntax error, because 09 is an incorrect literal value.

Question 51

Correct

Mark 2.00 out of 2.00

What is x after the following statements?

```
int x = 2;
int y = 1;
x *= y + 1;
```

Select one:

- A. x is 1.
- B. x is 2.
- C. x is 3.
- D. x is 4. ✔ (y + 1) is executed first and its result is multiplied with x and assigned to x.

(y + 1) is executed first and its result is multiplied with x and assigned to x.

(y + 1) is executed first and its result is multiplied with x and assigned to x.

The correct answer is:
x is 4.

Question 52

Correct

Mark 1.00 out of 1.00

Are the following four statements equivalent?

```
number += 1;
number = number + 1;
number++;
++number;
```

Select one:

- A. Yes ✔ See Table 2.5.
- B. No

See Table 2.5.

See Table 2.5.

The correct answer is: Yes

Question 53

Incorrect

Mark 0.00 out of 1.00

Which of the following is incorrect?

Select one:

- A. float x = 1.0;
- B. long x = 9; ✖
- C. int x = 9;
- D. double x = 1.0;

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The correct answer is: float x = 1.0;

Question 54

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
double x = 5.5;
int y = (int)x;
System.out.println("x is " + x + " and y is " + y);
```

Select one:

- A. x is 6 and y is 6
- B. x is 6.0 and y is 6.0
- C. x is 5.5 and y is 5 ✔ The value is x is not changed after the casting.
- D. x is 5.5 and y is 5.0
- E. x is 5 and y is 6

The value is x is not changed after the casting.

The value is x is not changed after the casting.

The correct answer is: x is 5.5 and y is 5

Question 55

Correct

Mark 1.00 out of 1.00

To obtain the current minute, use _____.

Select one:

- A. System.currentTimeMillis() / 1000 / 60 / 60 % 24
- B. System.currentTimeMillis() % 60
- C. System.currentTimeMillis() / 1000 / 60 % 60 ✔ See Listing 2.7.
- D. System.currentTimeMillis() / 1000 % 60
- E. System.currentTimeMillis() % 3600

See Listing 2.7.

See Listing 2.7.

The correct answer is: System.currentTimeMillis() / 1000 / 60 % 60

Question 56

Partially correct

Mark 0.33 out of 1.00

Which of the Boolean expressions below is incorrect? Please select all that apply.

Select one or more:

- A. (true) && (3 => 4)
- B. !(x > 0) && (x > 0)
- C. (x > 0) || (x < 0)
- D. (x != 0) || (x = 0)
- E. (-10 < x < 0) ✔ A: (3 => 4) should be (3 >= 4), D: (x = 0) should be (x == 0), and E: should be (-10 < x) && (x < 0)

A: (3 => 4) should be (3 >= 4), D: (x = 0) should be (x == 0), and E: should be (-10 < x) && (x < 0)

The correct answers are: (true) && (3 => 4), (x != 0) || (x = 0), (-10 < x < 0)

Question 57

Correct

Mark 1.00 out of 1.00

Which of the following assignment statements is illegal?

Select one:

- A. int t = 4.5; ✔ See the second paragraph in this section.
- B. short s = 10;
- C. float f = -34;
- D. int t = 23;

See the second paragraph in this section.

See the second paragraph in this section.

The correct answer is: int t = 4.5;

Question 58

Correct

Mark 1.00 out of 1.00

To obtain the current hour in UTC, use _____.

Select one:

- A. System.currentTimeMillis() % 3600
- B. System.currentTimeMillis() % 60
- C. System.currentTimeMillis() / 1000 % 60
- D. System.currentTimeMillis() / 1000 / 60 / 60 % 24 ✔ See Listing 2.7.
- E. System.currentTimeMillis() / 1000 / 60 % 60

See Listing 2.7.

See Listing 2.7.

The correct answer is: System.currentTimeMillis() / 1000 / 60 / 60 % 24

Question 59

Correct

Mark 2.00 out of 2.00

Assume x = 4 and y = 5, which of the following is true?

Select one:

- A. x < 5 && y < 5
- B. x < 5 || y < 5 ✔ x < 5 is true, but y < 5 is false. So A is false. B is true. C and D are both false, because x > 5 is false and y > 5 is false. The correct answer is B.
- C. x > 5 && y > 5
- D. x > 5 || y > 5

x < 5 is true, but y < 5 is false. So A is false. B is true. C and D are both false, because x > 5 is false and y > 5 is false. The correct answer is B.

x < 5 is true, but y < 5 is false. So A is false. B is true. C and D are both false, because x > 5 is false and y > 5 is false. The correct answer is B.

The correct answer is: x < 5 || y < 5

Question 60

Correct

Mark 1.00 out of 1.00

Which of the following expression results in a value 1?

Select one:

- A. 37 % 6 ✔ 2 % 1 is 0, 15 % 4 is 3, 25 % 5 is 0, and 37 % 6 is 1
- B. 2 % 1
- C. 25 % 5
- D. 15 % 4

2 % 1 is 0, 15 % 4 is 3, 25 % 5 is 0, and 37 % 6 is 1

2 % 1 is 0, 15 % 4 is 3, 25 % 5 is 0, and 37 % 6 is 1

The correct answer is: 37 % 6

Question 61

Correct

Mark 1.00 out of 1.00

Analyze the following code fragments that assign a boolean value to the variable even.

Code 1:
```
if (number % 2 == 0)
  even = true;
else
  even = false;
```

Code 2:
```
even = (number % 2 == 0) ? true: false;
```

Code 3:
```
even = number % 2 == 0;
```

Select one:

- A. Code 2 has a compile error, because you cannot have true and false literals in the conditional expression.
- B. Code 3 has a compile error, because you attempt to assign number to even.
- C. All three are correct, but Code 1 is preferred.
- D. All three are correct, but Code 2 is preferred.
- E. All three are correct, but Code 3 is preferred. ✔ Code 3 is the simplest. Code 1 and Code 2 contain redundant code.

Code 3 is the simplest. Code 1 and Code 2 contain redundant code.

Code 3 is the simplest. Code 1 and Code 2 contain redundant code.

The correct answer is: All three are correct, but Code 3 is preferred.

Question 62

Incorrect

Mark 0.00 out of 2.00

Analyze the following code:

```
boolean even = false;
if (even = true) {
  System.out.println("It is even");
}
```

Select one:

- A. The program has a runtime error.
- B. The program runs fine and displays It is even.
- C. The program has a compile error.
- D. The program runs fine, but displays nothing. ✖

It is a common mistake to use the = operator in the condition test. What happens is that true is assigned to even when you write even = true. So even is true. The program compiles and runs fine and displays "It is even".

The correct answer is: The program runs fine and displays It is even.

Question 63

Incorrect

Mark 0.00 out of 2.00

What is y displayed in the following code?

```
public class Test1 {
  public static void main(String[] args) {
    int x = 1;
    int y = x = x + 1;
    System.out.println("y is " + y);
  }
}
```

Select one:

- A. y is 2 because x + 1 is assigned to x and then x is assigned to y.
- B. y is 1 because x is assigned to y first.
- C. The program has a compile error since x is redeclared in the statement int y = x = x + 1. ✖
- D. y is 0.

The = operator is right-associative.

The correct answer is: y is 2 because x + 1 is assigned to x and then x is assigned to y.

Question 64

Correct

Mark 1.00 out of 1.00

What is the value of the following expression?
true || true && false

Select one:

- A. true ✔ && has higher precedence than ||, so && is evaluated first.
- B. false

&& has higher precedence than ||, so && is evaluated first.

&& has higher precedence than ||, so && is evaluated first.

The correct answer is: true

Question 65

Incorrect

Mark 0.00 out of 1.00

What is i printed in the following code?

```
public class Test {
  public static void main(String[] args) {
    int j = 0;
    int i = j++ + j * 5;

    System.out.println("What is i? " + i);
  }
}
```

Select one:

- A. 0 ✖
- B. 1
- C. 5
- D. 6

Operands are evaluated from left to right in Java. The left-hand operand of a binary operator is evaluated before any part of the right-hand operand is evaluated. This rule takes precedence over any other rules that govern expressions. Therefore, j++ is evaluated first. j is now 1. Since j++ is postincrement, the old value of j is returned for j++. So, j++ + j * 5 equals 0 + 1 * 5. So, the result is 5.

The correct answer is: 5

Question 66

Correct

Mark 1.00 out of 1.00

Which of the following assignment statements is incorrect? Please select all that apply.

Select one or more:

- A. i = 1 = j = 1 = k = 1; ✔ Read toward the end of the section.
- B. i = j = k = 1;
- C. i = 1; j = 1; k = 1;
- D. i == j == k == 1; ✔ Read toward the end of the section.

Read toward the end of the section.

Read toward the end of the section.

The correct answers are: i = 1 = j = 1 = k = 1;, i == j == k == 1;

Question 67

Correct

Mark 1.00 out of 1.00

Which of the following expression results in 45.37?

Select one:

- A. (int)(45.378 * 100) / 100.0 ✔ See Listing 2.8.
- B. (int)(45.378 * 100 / 100)
- C. (int)(45.378) * 100 / 100.0
- D. (int)(45.378 * 100) / 100

See Listing 2.8.

See Listing 2.8.

The correct answer is: (int)(45.378 * 100) / 100.0

Question 68

Correct

Mark 1.00 out of 1.00

Which of the following is a possible output from invoking Math.random()? Please select all that apply.

Select one or more:

- A. 3.43
- B. 1.0
- C. 0.5 ✔ Math.random() returns a real value between 0.0 and 1.0, excluding 1.0.
- D. 0.0 ✔ Math.random() returns a real value between 0.0 and 1.0, excluding 1.0.

Math.random() returns a real value between 0.0 and 1.0, excluding 1.0.

Math.random() returns a real value between 0.0 and 1.0, excluding 1.0.

The correct answers are: 0.5, 0.0

Question 69

Correct

Mark 1.00 out of 1.00

Which of the following is a constant, according to Java naming conventions? Please select all that apply.

Select one or more:

- A. COUNT ✔ All letters in a constant are in uppercase. In a multiple-word constant, the words are connected using underscores.
- B. ReadInt
- C. MAX_VALUE ✔ All letters in a constant are in uppercase. In a multiple-word constant, the words are connected using underscores.
- D. read ✖
- E. Test

All letters in a constant are in uppercase. In a multiple-word constant, the words are connected using underscores.

All letters in a constant are in uppercase. In a multiple-word constant, the words are connected using underscores.

The correct answers are: MAX_VALUE, COUNT

Question 70

Correct

Mark 1.00 out of 1.00

To assign a double variable d to a float variable x, you write _____.

Select one:

- A. x = (long)d
- B. x = (float)d; ✔ See the second paragraph in this section.
- C. x = (int)d;
- D. x = d;

See the second paragraph in this section.

See the second paragraph in this section.

The correct answer is: x = (float)d;

Question 71

Incorrect

Mark 0.00 out of 1.00

When assigning a literal to a variable of the byte type, if the literal is too large to be stored as a byte value, it _____.

Select one:

- A. causes underflow
- B. causes no error
- C. cannot happen in Java
- D. causes overflow ✖
- E. receives a compile error

For example, byte b = 23232 will cause a compile error.

The correct answer is: receives a compile error

Question 72

Correct

Mark 1.00 out of 1.00

Which of the following expressions will yield 0.5? Please select all that apply.

Select one or more:

- A. 1 / 2.0 ✔ 1 / 2 is an integer division, which results in 0.
- B. 1 / 2
- C. 1.0 / 2 ✔ 1 / 2 is an integer division, which results in 0.
- D. (double) 1 / 2 ✔ 1 / 2 is an integer division, which results in 0.
- E. (double) (1 / 2) ✖

1 / 2 is an integer division, which results in 0.

1 / 2 is an integer division, which results in 0.

The correct answers are: 1.0 / 2, (double) 1 / 2, 1 / 2.0

Question 73

Incorrect

Mark 0.00 out of 1.00

_____ seeks to analyze the data flow and to identify the system's input and output. When you do analysis, it helps to identify what the output is first, and then figure out what input data you need in order to produce the output.

Select one:

- A. Design
- B. Implementation
- C. Testing ✖
- D. Analysis
- E. Requirements specification

See the third paragraph in this section.

The correct answer is: Analysis

Question 74

Correct

Mark 2.00 out of 2.00

Assume x = 4, which of the following is true?

Select one:

- A. !(x == 4)
- B. x != 4
- C. x == 5
- D. x != 5 ✔ D is true. All others are false.

D is true. All others are false.

D is true. All others are false.

The correct answer is: x != 5

Question 75

Correct

Mark 2.00 out of 2.00

What is y after the following statement is executed?

```
x = 0;
y = (x > 0) ? 10 : -10;
```

Select one:

- A. -10 ✔ This conditional operator is correct. It assigns -10 to y since x > 0 is false.
- B. 0
- C. 10
- D. 20
- E. Illegal expression

This conditional operator is correct. It assigns -10 to y since x > 0 is false.

This conditional operator is correct. It assigns -10 to y since x > 0 is false.

The correct answer is: -10